Current approaches to modelling the **spread** of **wildland fire**: a **review**

GLW Perry - Progress in Physical Geography, 1998 - journals.sagepub.com

… **review** considers the development of some of the models and modelling approaches designed to predict the **spread** and spatial behaviour of **wildland fire** … for **wildland fire management** …

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Machine learning and deep learning for **wildfire spread prediction**: A **review**

HS Andrianarivony, MA Akhloufi - Fire, 2024 - mdpi.com

… highly performant in predicting fire **behavior**. This paper presents a systematic **review** of recent ML and DL techniques developed for **wildfire spread prediction**, detailing the commonly …

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Trending and emerging prospects of physics-based and ML-based **wildfire spread** models: A comprehensive **review**

H Singh, LM Ang, T Lewis, D Paudyal, M Acuna… - Journal of Forestry …, 2024 - Springer

… aiding in the development of more accurate and reliable **prediction** tools. Methods for literature **review** This **wildfire spread model review** analysed research published in peer-reviewed …

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A comparative **review** on **wildfire** simulators

GD Papadopoulos, FN Pavlidou - IEEE systems Journal, 2011 - ieeexplore.ieee.org

… This project promises to develop more reliable **bushfire spread** simulation and animation … In this paper, a **review** on **wildfire simulation** models has been presented. Before that, a brief …

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Uncertainty and risk in **wildland fire management**: a **review**

MP Thompson, DE Calkin - Journal of environmental **management**, 2011 - Elsevier

… into the impacts of wildfires on human life and property, ecosystems, and other valued resources, and into the factors contributing to **wildfire** occurrence and **spread**. Managing …

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A systematic **review** and bibliometric analysis of **wildland fire behavior** modeling

J Silva, J Marques, I Gonçalves, R Brito, S Teixeira… - Fluids, 2022 - mdpi.com

… a mathematical percept to **model** the **spread** of **wildland fire**. Bakhshaii and Johnson [7], in a recent **review**, called these last two models a first-generation type of **wildfire** models. The …

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A **review** of a new generation of **wildfire**–atmosphere modeling

A Bakhshaii, EA Johnson - Canadian Journal of Forest …, 2019 - cdnsciencepub.com

… The earliest physical science models tend to deal with isolated microscale phenomena (eg, **wildfire spread model** or the optical properties of ice crystals) so that their direct application …

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A **review** of **wildland fire spread** modelling, 1990-present 3: Mathematical analogues and **simulation** models

AL Sullivan - arXiv preprint arXiv:0706.4130, 2007 - arxiv.org

… involving **wildland fire spread** that are not based on **wildland fire** … to the **spread** of fires and thus are described as **wildland fire spread** models, while in some cases **wildland fire spread** is …

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A **review** of machine learning applications in **wildfire** science and **management**

P Jain, SCP Coogan, SG Subramanian… - Environmental …, 2020 - cdnsciencepub.com

… For example, in the task of modeling **wildland fire spread**, the relevant physical properties include fuel composition, local weather, and topography. The current state-of-the-art method in …

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A Comprehensive **Review** of Empirical and Dynamic **Wildfire** Simulators and Machine Learning Techniques used for the **Prediction** of **Wildfire** in Australia

H Singh, LM Ang, D Paudyal, M Acuna… - Technology, Knowledge …, 2025 - Springer

… better anticipate and mitigate the impacts of **wildfire spread**. This **review** will discuss the range of empirical and dynamic tools used for **wildfire prediction**, highlighting their advantages, …

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A **review** of recent advances in risk analysis for **wildfire management**

C Miller, AA Ager - International journal of **wildland fire**, 2012 - CSIRO Publishing

… We also **review** recent advances in addressing temporal **dynamics** of fire risk and spatial optimisation of fuels **management** … for future research and development in **wildfire** risk analysis. …

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A **review** of **wildland fire spread** modelling, 1990-present, 1: Physical and quasi-physical models

AL Sullivan - arXiv preprint arXiv:0706.3074, 2007 - arxiv.org

… Whereas FIRETEC was intended to be used to **model wildland fire spread** across large spatial scales, IUSTI concentrates on resolving the chemical and conservation equations at a …

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Mathematical models and calculation systems for the study of **wildland fire** behaviour

E Pastor, L Zárate, E Planas, J Arnaldos - Progress in Energy and …, 2003 - Elsevier

This is a **review** of the most important work in **wildland fire** mathematical modelling which … **wildland fire** models to be sorted. Surface fire **spread** models, crown fire initiation and **spread** …

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Operational **wildfire** suppression modelling: a **review** evaluating development, state of the art and future directions

TJ Duff, KG Tolhurst - International Journal of **Wildland Fire**, 2015 - CSIRO Publishing

… **review**, we focus on the development and use of models that emulate fire suppression for decision support. We **review** … There are two main approaches to dynamic fire **spread simulation**…

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Wildland surface fire **spread** modelling, 1990–2007. 3: **Simulation** and mathematical analogue models

AL Sullivan - International Journal of **Wildland Fire**, 2009 - CSIRO Publishing

… The present series of articles endeavours to comprehensively **survey** and précis all types of surface fire **spread** models developed during the period 1990–2007. The present paper …

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Modelling smoke transport from wildland fires: a **review**

SL Goodrick, GL Achtemeier, NK Larkin… - … of **Wildland Fire**, 2012 - CSIRO Publishing

… Another grid **model** receiving attention for **wildland fire** related air quality issues is the … of the modelling studies cited in this **review** involves the **dynamics** of the buoyant phase of the …

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Limitations on the accuracy of **model** predictions of **wildland fire** behaviour: a state-of-the-knowledge overview

ME Alexander, MG Cruz - The forestry chronicle, 2013 - pubs.cif-ifc.org

… **model** predictions of **wildland fire** behaviour and includes a short historical overview of **wildland fire** behaviour research as it relates to **model** … Estimating **wildland fire** rate of **spread** in a …

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DEVS-FIRE: Towards an integrated **simulation** environment for surface **wildfire spread** and containment

L Ntaimo, X Hu, Y Sun - **Simulation**, 2008 - journals.sagepub.com

… applied to an example of fire **spread simulation** [44]. A recent … the **wildfire spread** and containment **simulation model** presented in … We **review** the basics of **wildfire behavior** [56] to set …

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Wildfire management in Canada: **Review**, challenges and opportunities

C Tymstra, BJ Stocks, X Cai, MD Flannigan - Progress in Disaster Science, 2020 - Elsevier

… **management** agencies to allow more **wildfire** on the landscape by implementing a risk-based … future path for **wildfire management** in Canada. To co-exist with **wildfire**, agencies in …

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Wildland surface fire **spread** modelling, 1990–2007. 1: Physical and quasi-physical models

AL Sullivan - International Journal of **Wildland Fire**, 2009 - CSIRO Publishing

… readers, and this is the main reason for the inclusion of his work in the current **review**. … Whereas FIRETEC was intended to be used to **model wildland fire spread** across large spatial …

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